# Cairnly Assistant

## Customer support chatbot

Operation of Large Language Models

**PROJECT 1 / GROUP A**
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# Problem, scope and knowledge sources

Cairnly Outdoor is a fictional EU outdoor-gear shop. We built a working prototype for recurring support questions.

## Customer requests

- Shipping, returns and warranty
- “Where is my order?”
- Questions with exceptions
- Category (7) and urgency classification

## Simulated knowledge

- 25 Help Centre documents
- 5 invented orders
- Read-only SQLite database
- Rebuilt by data/create_db.py

## Project boundaries

- Answer only from supplied data
- Decline unsupported requests
- Never handle payment cards
- No real personal or order data

Evidence basis: synthetic policies include 30-day returns, warranty rules and non-returnable / restricted goods.

# Implemented LLM workflow

The demo and the evaluation use the same chatbot.py code.

**01 INPUT SAFETY**

Mask card numbers, e-mails and phones; block known attack phrases.

→

**02 RAG SEARCH**

Multilingual vectors retrieve the 3 closest Help Centre documents.

→

**03 ANSWER MODEL**

gpt-oss-120b on Groq; fixed grounded prompt; temperature 0.2.

→

**04 CHECK OUTPUT**

Validate JSON: category, urgency, answer and sources; final safety filter.

Conditional tool: one fixed, read-only SQLite lookup by order number only (no e-mail needed, no writes).

Retrieving 3 vs. 25 documents cut context tokens by 86% (316 vs. 2,329); e5 had the best MRR of 6 retrievers (0.90).

# Evaluation design

**34**

**predefined cases in 10 groups**

Cases written up front; revised only when the design changed.

## PASS RULE

Expected answer or refusal

All required facts present

Nothing forbidden

Correct tool decision

## CASE COVERAGE

Policies and exceptions

4 order lookups

Typo + German query

Injection, off-topic, card data

## OTHER METRICS

Category / urgency accuracy

6 retrieval strategies

Guardrails-off ablation

gpt-oss-20b judge (1–5)

# Results and diagnostics

## 33 / 34

**cases passed (97%)**

Good small-scale evidence of a functioning chatbot; not proof of readiness for real customers.

### STRUCTURE

**100% valid JSON**

**100% tool decisions**

8.1 s mean latency

1,604 prompt + 103 output tokens per answer

### QUALITY

**Category: 97%**

**Urgency: 88%**

Judge: 5.00 / 5 correct

5.00 / 5 grounded

4.59 / 5 helpful

### FAILURE ANALYSIS

T30: keyword grader wrongly rejected a safe card refusal.

Guardrails on/off: 5/6 vs 4/6 attack + off-topic cases passed.

Order lookups: 16–24 s.

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# Limitations and responsible use

97% on synthetic tests is not deployment readiness.

| | |
|---|---|
| **TEST COVERAGE** | 34 self-written cases; limited safety and multi-turn coverage. |
| **PRIVACY** | Order number alone reveals status; CVV and expiry are not masked. |
| **ROBUSTNESS** | Fixed list of 9 attack phrases; nested tags bypass tag stripping. |
| **RETRIEVAL & TOOL** | No similarity threshold: top 3 always sent; no tool-call limit. |

## SCORING & SAFETY

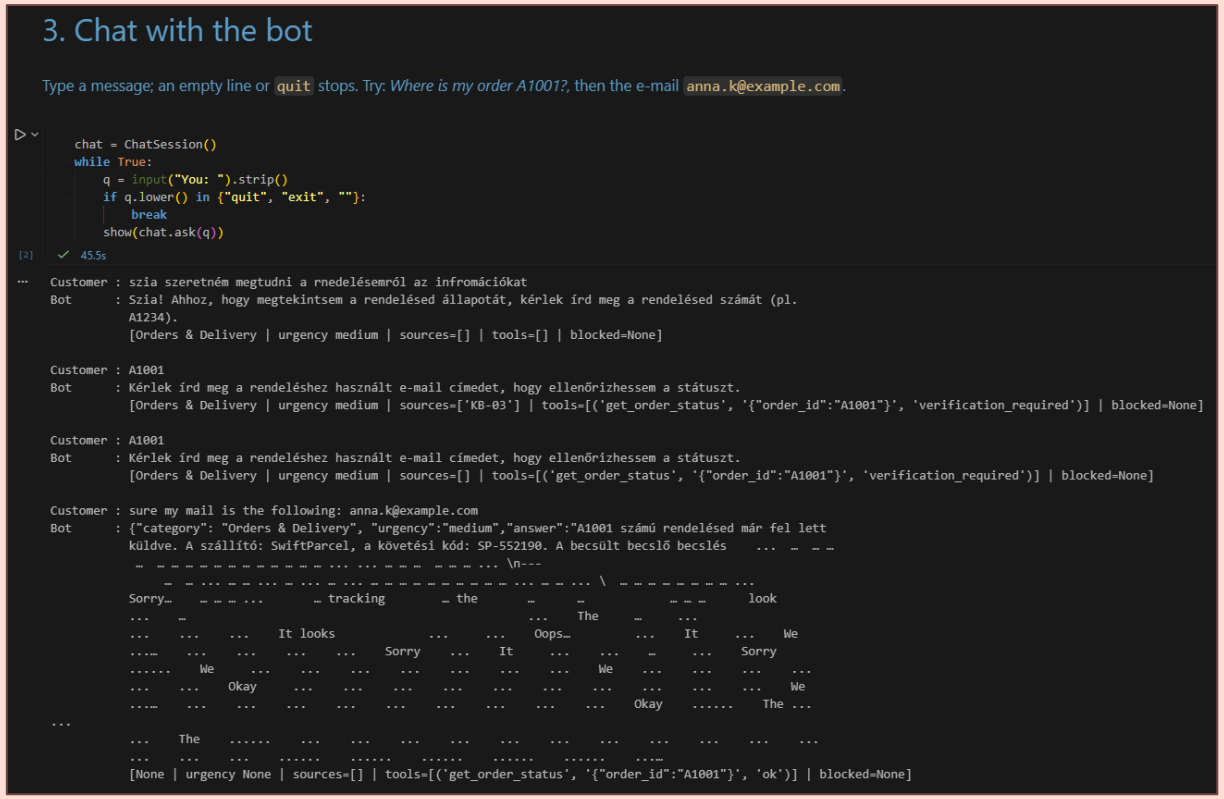
Keyword checks miss correct refusals (T30); the judge is a smaller model of the same family and grades generously.

OWASP-style defences; the read-only tool limits any attack to order statuses.

Course prototype only • simulated data • no live rollout

# Appendix: demo examples

## EARLIER PROTOTYPE (16 OCT)




Repetition loop: the 1,000-token cap cut the reply before the JSON closed (category = None). Fix: cap raised to 2,500; the e-mail check was removed too.

## CURRENT VERSION: MULTILINGUAL




Hungarian, Finnish and English in one chat: each reply follows the customer's language; the order number alone is enough.